

ENTHUSIAST, LADER & ACHIEVER COURSE

PHASE : (All Phase)

TARGET : PRE-MEDICAL 2020

Test Type : DRILL TEST # 07

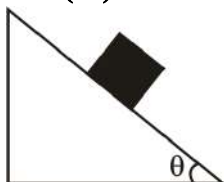
Test Pattern : NEET (UG)

TEST DATE : 13-04-2020

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A.	2	3	2	4	4	2	4	2	4	4	2	2	3	3	1	2	2	1	3	4	3	1	1	1	3	4	2	2	1	1
Q.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
A.	1	3	4	2	2	1	3	4	3	3	4	2	1	3	2	4	4	1	3	2	2	2	1	2	4	4	1	2	1	2
Q.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
A.	1	3	2	4	1	3	4	4	3	2	4	2	4	1	3	1	4	4	2	1	2	1	1	1	4	3	2	4	2	2
Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
A.	2	4	3	2	1	2	2	4	3	4	3	4	3	3	3	3	4	3	2	3	2	3	2	3	3	4	3	3	2	3
Q.	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	3	4	1	2	4	2	1	1	4	2	3	3	1	2	2	4	2	2	2	2	3	4	1	4	1	3	1	4	1	2
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180
A.	2	1	4	1	3	3	2	1	3	3	4	3	2	4	4	2	3	4	2	4	1	3	2	1	4	3	2	3	4	2

HINT - SHEET

1. Ans (2)



$$\begin{aligned}
 &F_{cp} = N \sin \theta + f \cos \theta \quad \dots (1) \\
 &Mg + f \sin \theta = N \cos \theta \quad \dots (2)
 \end{aligned}$$

2. Ans (3)

$$V_{\text{(inside sphere)}} = V_{\text{surface}}$$

3. Ans (2)

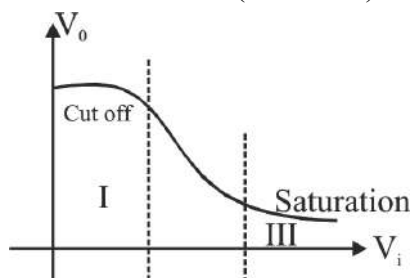
$$\begin{aligned}
 H_{\text{max}} &= 60 + H^1, H^1 = \frac{u^2}{2g} = 20 \text{ m} \\
 \text{then } &= 60 + 20 = 80
 \end{aligned}$$

4. Ans (4)

$$\begin{aligned}
 P_0 V_0 &= nRT_0 \\
 (3P_0)(3V_0) &= nRT_B \\
 9P_0 V_0 &= nRT_B \\
 9nRT_0 &= nRT_B \Rightarrow T_B = 9T_0
 \end{aligned}$$

5. Ans (4)

As a switch In saturation (on switch) → In cut off (off switch)



6. Ans (2)

$$i_{\text{ind}} = \frac{Bv_{\text{max}} \ell}{R}$$

7. Ans (4)

$$\frac{L}{K \cdot E} = \frac{mVR_0}{\frac{1}{2}mV^2} = \frac{2R_0}{V} = 2\sqrt{\frac{R_0^3}{GM}}$$

8. Ans (2)

$$\text{Formula } R = \frac{\ell}{KA}$$

9. Ans (4)

$$B \text{ \& C are in } \parallel \ell \text{ \& } R_A = R_B \parallel R_C$$

10. **Ans (4)**

$$Y = \overline{AB} + \overline{AB} = (\overline{AB}) \cdot (\overline{AB}) = 0$$

11. **Ans (2)**

In case of diffraction at single slit, the position of minima is given by

$$d \sin \theta = n\lambda \quad \dots(1)$$

$$\text{and for small } \theta : \sin \theta = \theta = (y/D) \quad \dots(2)$$

So, from Eqs. (1) and (2), we have :

$$d \left(\frac{y}{D} \right) = n\lambda, \text{ i.e. } y = \frac{D}{d}(n\lambda)$$

$$\text{So that, } y_3 - y_1 = \frac{D}{d}(3\lambda - \lambda) = \frac{D}{d}(2\lambda)$$

$$\text{and hence, } d = \frac{0.50 \times (2 \times 6 \times 10^{-7})}{3 \times 10^{-3}}$$

$$= 2 \times 10^{-4} \text{ m}$$

$$= 0.2 \text{ mm}$$

12. **Ans (2)**

at resonance power is max so $X_L = X_C$

13. **Ans (3)**

$$\langle v \rangle = \frac{\int_0^1 (2t + 3) dt}{\int_0^1 dt} = \frac{[t^2 + 3t]_0^1}{[1 - 0]} = 4 \text{ m/s}$$

14. **Ans (3)**

Potential energy of the spring when stretched through a distance x .

$$U = \frac{1}{2} kx^2 = 10 \text{ J}$$

When x becomes $2x$, the potential energy will be

$$U' = \frac{1}{2} k(2x)^2 = 4 \times \frac{1}{2} kx^2 = 4 \times 10 = 40 \text{ J}$$

$$\therefore \text{Work done } U' - U = 40 - 10 = 30 \text{ J}$$

15. **Ans (1)**

Comparing given equation with standard equation

$$y = 2a \sin \frac{2\pi x}{\lambda} \cos \frac{2\pi vt}{\lambda} \text{ gives us}$$

$$\frac{2\pi}{\lambda} = \frac{\pi}{15} \Rightarrow \lambda = 30$$

Distance between nearest node and antinodes.

$$= \frac{\lambda}{4} = \frac{30}{4} = 7.5$$

16. **Ans (2)**

Acceleration of a purely rolling object on an inclined plane is :-

$$a = \frac{g \sin \theta}{\left(1 + \frac{K^2}{R^2}\right)}$$

$$\text{for spherical shell, } \frac{K^2}{R^2} = \frac{2}{3}$$

$$\text{for solid cylinder, } \frac{K^2}{R^2} = \frac{1}{2}$$

$$\text{so, } \frac{a_{\text{shell}}}{a_{\text{cylinder}}} = \frac{\frac{g \sin \theta}{\left(1 + \frac{2}{3}\right)}}{\frac{g \sin \theta}{\left(1 + \frac{1}{2}\right)}} = \frac{9}{10}$$

17. **Ans (2)**

$$\vec{F}_1 + \vec{F}_2 + \vec{F}_3 + \vec{F}_4 + \vec{F}_5 = 2(4\hat{i}) = 8\hat{i}$$

$$\vec{F}_2 + \vec{F}_3 + \vec{F}_4 + \vec{F}_5 = 2(7\hat{j}) = 14\hat{j}$$

$$\vec{F}_1 + 14\hat{j} = 8\hat{i}$$

$$\vec{F}_1 = 8\hat{i} - 14\hat{j}$$

$$F_1 = \sqrt{(8)^2 + (14)^2} = \sqrt{260} = 2\sqrt{65}$$

$$\frac{F_1}{m} = \frac{2\sqrt{65}}{2} = \sqrt{65} \text{ m/s}^2$$

18. **Ans (1)**

$$\Delta P\% = \frac{P_f - P_i}{P_i} \times 100$$

$$= \left(\frac{P_f}{P_i} - 1 \right) \times 100$$

$$= \left(\sqrt{\frac{K \cdot E_f}{K \cdot E_i}} - 1 \right) \times 100$$

$$= (\sqrt{3} - 1) \times 100$$

$$= 73.2 \%$$

19. **Ans (3)**

$$1 - \frac{T_2}{T_1} = \frac{1}{5} \Rightarrow \frac{T_2}{T_1} = \frac{4}{5}$$

$$1 - \frac{(T_2 - 50)}{T_1} = \frac{1}{3} \Rightarrow T_1 = 375 \text{ K}$$

$$T_2 = \frac{4}{5} (375) = 300 \text{ K}$$

20. **Ans (4)**

$$MP_{\min} = \frac{D}{f} = \frac{25}{12.5} = 2$$

$$MP_{\max} = \frac{D}{f} + 1 = \frac{25}{12.5} + 1 = 3$$

21. **Ans (3)**

electrostatic field lines do not form closed loop

22. **Ans (1)**

$$F = IBL$$

23. **Ans (1)**

$$I = I_{CM} + Md^2$$

24. **Ans (1)**

$$m\vec{u}_1 = m_1\vec{v}_1 + m_2\vec{v}_2$$

$$mv_0\hat{i} = \frac{m}{3}(2v_0\hat{j}) + \frac{2m}{3}\vec{v}_2$$

$$\vec{v}_2 = \frac{3}{2}v_0\hat{i} - v_0\hat{j}$$

25. **Ans (3)**

$$19 \times 10^{10} \frac{N}{m^2} = 19 \times 10^{10} \times 10^5 \frac{\text{dyne}}{10^4 \text{cm}^2}$$

26. **Ans (4)**

Conceptual

27. **Ans (2)**

$$\frac{2\pi}{T} = 100\pi$$

$$T = .02 \text{ sec}$$

28. **Ans (2)**

$$V_a = -\frac{GM}{a/2} - \frac{GM}{a} = -\frac{3GM}{a}$$

29. **Ans (1)**

$$0.18v + 0.12 \times 1.5 = 0.12 \times 3$$

$$\text{or } 0.18v = 0.12 \times 1.5$$

$$\text{or } v = \frac{0.12 \times 15}{0.18} \text{ms}^{-1} = 1 \text{ms}^{-1}$$

30. **Ans (1)**

$\angle$ between M & B is 30° so between B & plane of coil = 60°

31. **Ans (1)**

$$\text{voltage on } 40 \text{ cm} = 4.8 \text{ V}$$

$$\text{Potential gradient} = \frac{4.8}{40 \text{cm}}$$

$$\text{voltage on } 60 \text{ cm so} = \frac{4.8}{40} \times 60 = 7.2 \text{ volt.}$$

32. **Ans (3)**

$$r = \frac{r_1 r_2}{r_2 - r_1} = \infty (r_1 = r_2)$$

34. **Ans (2)**

$$\frac{hc}{\lambda_A} = K_A + W$$

$$\Rightarrow \frac{hc}{2\lambda_B} = K_A + W$$

$$\Rightarrow \frac{hc}{\lambda_B} = 2K_A + 2W \quad \dots(1)$$

$$\text{and } \frac{hc}{\lambda_B} = K_B + W \quad \dots(2)$$

by (1) and (2)

$$2K_A + 2W = K_B + W$$

$$\Rightarrow \frac{K_B}{2} - K_A = \frac{W}{2}$$

35. **Ans (2)**

$$N = \frac{N_0}{2^2} - \frac{N_0}{2^4}$$

36. **Ans (1)**

$$V = 5 \text{ km/hr}$$

$$d = 1 \text{ km}$$

$$t = 15 \text{ min} = \frac{1}{4} \text{ hrs}$$

$$V \sin \theta = V_R t = \frac{d}{V \cos \theta}, \cos \theta = \frac{d}{tV}$$

$$\cos \theta = \frac{1}{\frac{1}{4} \cdot 5} = \frac{4}{5}, \theta = 37^\circ$$

$$V_R = V \sin \theta = 5 \sin 37^\circ$$

$$V_R = 5 \cdot \frac{3}{5} = 3 \text{ km/hr}$$

37. **Ans (3)**

Fe has highest work function

38. **Ans (4)**

$$E_y = 2.5 \frac{N}{C} \cos[(2\pi \times 10^6 t) - \pi \times 10^{-2} x]$$

direction - X axis

$$\text{wavelength } \lambda = \frac{2\pi}{k} = \frac{2\pi}{\pi \times 10^{-2}} = 200 \text{ m}$$

$$f = \frac{\omega}{2\pi} = \frac{2\pi \times 10^6}{2\pi} = 10^6 \text{ Hz}$$

39. **Ans (3)**

$$Q = BE_{(\text{prod})} - BE_{(\text{React})}$$

40. **Ans (3)**

$$W = \frac{\pi}{2} (20) (20) = 200 \pi \text{ J}$$

41. Ans (4)

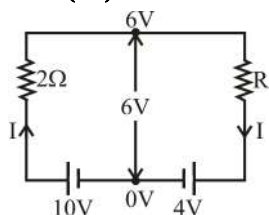
$$v = \frac{\Delta\lambda}{\lambda} C = \frac{1.9}{5700} \times 3 \times 10^8 = 1 \times 10^5 \text{ m/s}$$

42. Ans (2)

$$n_0 \sin \theta = \frac{n_0}{8} \sin 90^\circ$$

$$\sin \theta = \frac{1}{8}$$

43. Ans (1)



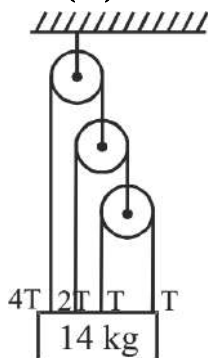
$$6 = 10 - 2I \Rightarrow I = 2 \text{ Amp.}$$

$$\text{Also } 6 = 4 + IR$$

$$6 = 4 + 2(R)$$

$$\Rightarrow R = 1\Omega$$

44. Ans (3)



$$\text{Therefore, } 4T + 2T + T + T = 14g$$

$$8T = 140$$

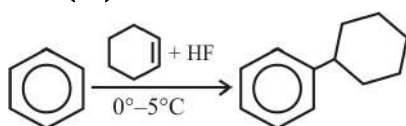
= Tension in string connecting ceiling and topmost pulley

45. Ans (2)

$$\tau_1 = RC \text{ and } \tau = 2RC$$

$$\Rightarrow \frac{\tau_1}{\tau_2} = \frac{1}{2}$$

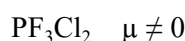
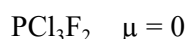
47. Ans (4)



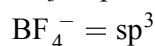
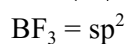
48. Ans (1)

Ma_4b_2 shows G.I.

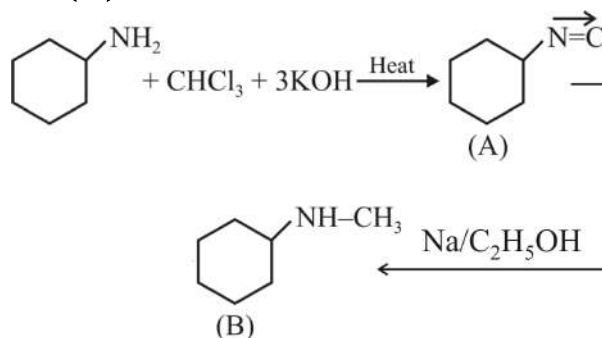
49. Ans (3)



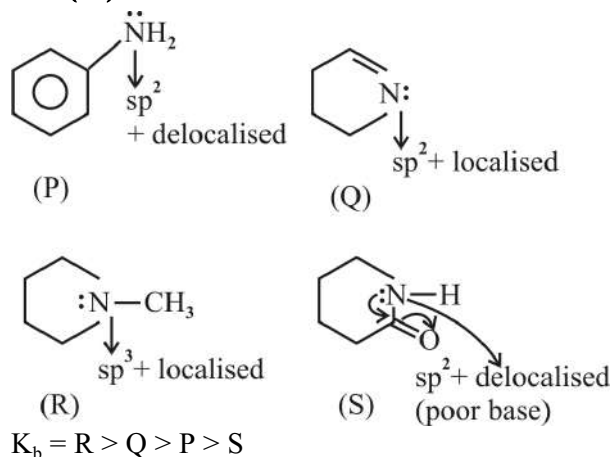
50. Ans (2)



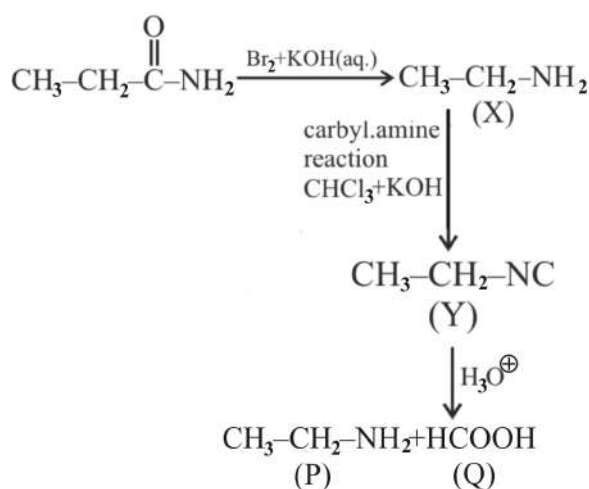
51. Ans (2)



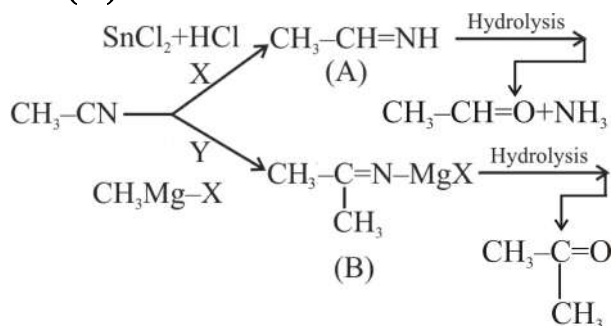
52. Ans (2)



53. Ans (1)



54. Ans (2)



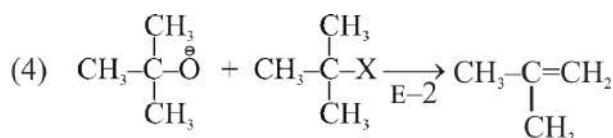
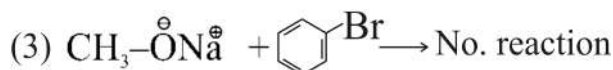
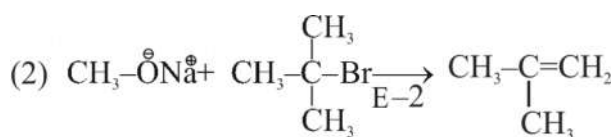
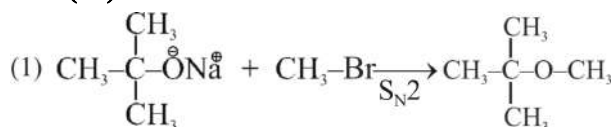
55. Ans (4)

Fact

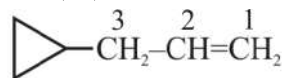
56. Ans (4)

Potassium ferricyanide does not give the test of ferric ion as it is available as $[\text{Fe}(\text{CN})_6]^{-3}$ complex ion.

57. Ans (1)



58. Ans (2)



3-Cyclopropyl-1-propene

59. Ans (1)

$$\Delta_B^\circ - \Delta_C^\circ = 20$$

$$\Delta_{\text{CX}}^\circ = \Delta_{\text{BX}}^\circ - 20$$

$$= 198 - 20$$

$$= 178$$

62. Ans (3)

$$K_p < K_c \text{ if } \Delta n_g < 0$$



63. Ans (2)

$$A_t = A_0 - kt$$

$$A_t = 0.1 - 2 \times 10^{-2} \times 2$$

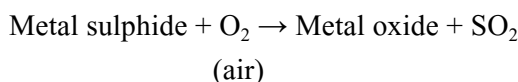
$$= 0.1 - 0.04$$

$$= 0.06 \text{ mol l}^{-1}$$

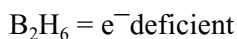
64. Ans (4)

Ozonolysis is a oxidative / Reductive cleavage of $>\text{C}=\text{C}<$ and $-\text{C}\equiv\text{C}-$

66. Ans (3)

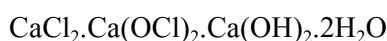


67. Ans (4)



68. Ans (4)

Bleaching powder



70. Ans (2)

$$\Delta E = q + w$$

$$q = 150 \text{ cal} = 150 \times 4.2 = 630 \text{ J}$$

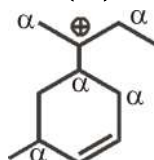
$$W = 200 \text{ J}$$

$$\Delta E = q + w$$

$$= 630 + 200$$

$$= 830 \text{ J}$$

72. Ans (2)



No. of $\alpha\text{-C-H}$ bond at sp^3

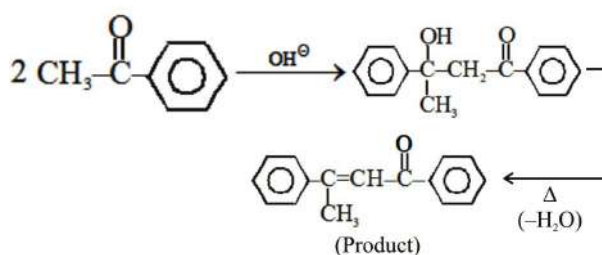
hybridised $\alpha\text{-C-atom} = 9$

74. Ans (1)

0.25 moles of urea is present in 1000 gm

$$m_{\text{urea}} = \frac{0.25 \times 60}{1015} \times 2500 = 37 \text{ g}$$

75. Ans (3)

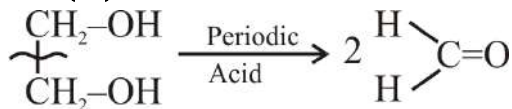


76. Ans (1)

$$K = A \cdot e^{-\frac{E_a}{RT}}$$

$$T \uparrow \Rightarrow K \uparrow$$

80. Ans (1)



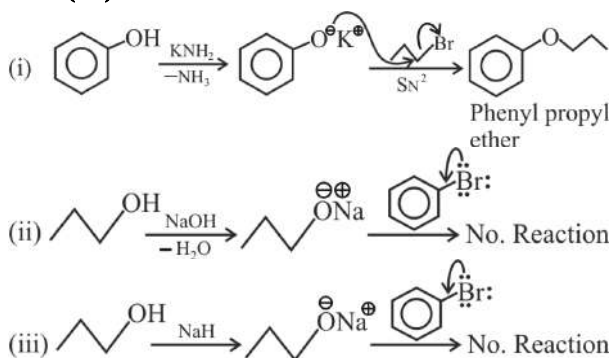
81. Ans (2)

$$\begin{aligned} \text{moles of 'O' atoms} &= \text{moles of } (\text{COOH})_2 \cdot 2\text{H}_2\text{O} \times 6 \\ &= 0.3 \times 6 \\ &= 1.8 \text{ mole} \\ &= 1.8 \text{ gm-atom} \end{aligned}$$

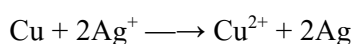
82. Ans (1)

Both are positional isomers.

83. Ans (1)



84. Ans (1)



$$Q = \frac{10^{-2}}{(10^{-1})^2} = 1$$

$$E = 0.46 - \frac{0.06}{2} \log 1$$

$$E = 0.46 \text{ V}$$

86. Ans (3)

$$X_A = \frac{1}{2}, X_B = \frac{1}{2}$$

$$P_T = P_A^\circ X_A + P_B^\circ X_B$$

$$\begin{aligned} P_T &= 80 \times \frac{1}{2} + 60 \times \frac{1}{2} \\ &= 70 \text{ mm of Hg} \end{aligned}$$

87. Ans (2)

$$\Delta H = \Delta E + \Delta n_g RT$$

$$\Delta H = 2.1 \times 10^3 + 2 \times 2 \times 300$$

$$= 2100 + 1200$$

$$= 3300$$

$$\Delta G = \Delta H - T\Delta S = 3300 - 300 \times 20 = -2700 \text{ cal}$$

$$= -2.7 \text{ K cal}$$

88. Ans (4)

$$m_B = \frac{m_A}{2}$$

$$V_B = \frac{V_A}{4}$$

$$\frac{\lambda_B}{\lambda_A} = \frac{m_A V_A}{m_B V_B}$$

$$\lambda_B = \lambda_A \left(\frac{m_A V_A}{m_A \times V_A} \times 2 \times 4 \right)$$

$$\boxed{\lambda_B = 8\lambda}$$

89. Ans (2)

$$\frac{V_1}{T_1} = \frac{V_2}{T_2}$$

$$\Rightarrow \frac{400}{300} = \frac{V_2}{270}$$

$$V_2 = 360 \text{ ml}$$

$$\Delta V = \frac{40}{400} \times 100 = 10\%$$

103. Ans (3)

NCERT XII, Pg.#177

109. Ans (2)

NCERT Pg.# 66

110. Ans (3)

NCERT (XIth) Pg. # 305

112. Ans (3)

NCERT-XI Bio(E) Pg. # 116-Para No.-2, Line-9

125. Ans (4)

NCERT-XI Pg. # 261 Fig. 16.6

128. Ans (1)

NCERT XI Page # 307

131. Ans (3)

Module Pg. # 124

136. **Ans (4)**

NCERT (XI) Pg. # 136, para-02

140. **Ans (2)**

NCERT-XII, Page # 214 (12.4)

144. **Ans (4)**

NCERT (XIIth) Pg. # 196 (Para 11.1)

149. **Ans (1)**

NCERT (XII) Pg. # 259, para-15.1.1

161. **Ans (4)**

NCERT (XII) Pg. # 187

165. **Ans (4)**

Module-4, Pg. # 216,217

180. **Ans (2)**

NCERT XIIth Pg. # 169, para 9.1.4